

MARIZFIRST

App Flow Document

Healthcare Appointment Discovery + Assisted Booking Platform

Document Purpose

This App Flow Document defines every core user journey, screen transition, system state, permission gate, edge case, notification trigger, analytics event, and operational handoff required to build MarizFirst as a production appointment assistance platform. It translates the PRD/TRD into implementable user and system flows.

Item	Detail
Product	MarizFirst
Document type	App Flow Document
Version	AFD v1.0
Source baseline	MarizFirst PRD v1.0 and TRD v1.0
MVP model	Hybrid booking: patient selects a preferred slot; appointment remains Pending Confirmation until admin or partner approval.
Core users	Patients, doctors, hospitals, clinics, MarizFirst admin team, content/SEO team.
Positioning rule	Do not position the platform as a discount or coupon product. Focus on trusted healthcare discovery, verified profiles, clear appointment status, and assisted confirmation.

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1. Flow Design Principles

The MarizFirst app flow must be designed around trust, operational control, and fast patient action. The product is not a simple listing website. It is an assisted booking platform where every appointment request moves through a controlled confirmation workflow.

Principle	Flow Requirement	Why It Matters
Hybrid confirmation	Never show public auto-confirmation at MVP. Every booking starts as Pending Confirmation.	Protects trust when clinics/doctors change availability manually.
Short patient path	Search -> profile -> slot -> short form -> WhatsApp acknowledgement.	Reduces drop-off and improves appointment request conversion.
Operational visibility	Every pending booking must appear in admin and partner flows with status, source, slot, and SLA context.	The business depends on appointment operations, not only frontend traffic.
Scoped partner access	Partners only see their own profiles, slots, and appointment requests.	Prevents privacy leaks and partner trust issues.
SEO-first discovery	Doctor, hospital, clinic, city, speciality, and blog pages must be indexable and internally linked.	Organic discovery is a core acquisition channel.
Privacy-first analytics	Track funnel events without sending phone, email, or health concern text to public analytics.	Protects patient-sensitive data.
No dead ends	Every empty, failed, or blocked flow must offer a next best action.	Keeps users moving toward WhatsApp support or a different appointment option.

2. Role Map and Entry Points

Role	Primary Entry Point	Core Goal	Allowed Flow
Patient/Public user	Home, SEO page, listing, profile, blog, WhatsApp CTA	Find trusted care and request appointment quickly.	Browse public pages, search, view profiles, submit appointment request, receive updates.
Doctor partner	Partner login -> appointments/slots/profile	Receive and manage scoped appointment requests.	Manage own profile fields, slots, accept/reject/reschedule requests.
Hospital partner	Partner login -> organization appointments/linked doctors	Acquire patients across departments and manage requests.	Manage hospital profile, linked doctors, scoped appointments, slots where applicable.
Clinic partner	Partner login -> clinic profile/slots/appointments	Increase local appointment demand.	Manage clinic profile, slots, scoped appointment requests.
Admin	Admin login -> dashboard/appointment inbox	Control operations, listings, partners, and confirmations.	Manage all appointments, profiles, partners, content, SEO, audit logs, notifications.
Content/SEO editor	Admin login -> content/SEO sections	Publish safe and optimized content.	Manage blogs, FAQs, metadata, city/speciality copy. No patient or finance data.
Super admin	Admin login -> system/users/settings	Own system-level controls.	Full access with extra confirmation for dangerous actions.

3. Public App Navigation Flow

Primary Navigation Logic

Public navigation must support quick healthcare discovery while protecting SEO. Every core page should lead users toward profile views and appointment requests without forcing login.

Entry Source	User Intent	Next Screen	Primary CTA	Secondary CTA
Home page	Search for care by location, speciality,	Search results/listing or profile page.	Book Appointment	Talk on WhatsApp

Entry Source	User Intent	Next Screen	Primary CTA	Secondary CTA
Doctors listing	doctor, hospital, or clinic. Compare doctors and filter by city/speciality/availability.	Doctor profile.	View Profile	Book Appointment
Doctor SEO page	Find a doctor in a city/speciality.	Doctor profile or filtered listing.	Book Appointment	WhatsApp Support
Hospital listing/profile	Explore facility, departments, linked doctors, address.	Hospital profile or linked doctor page.	Request Appointment	Call/WhatsApp Support
Clinic listing/profile	Find local clinic or team.	Clinic profile/booking form.	Request Appointment	WhatsApp Support
Blog article	Learn and move to relevant care pathway.	Speciality/city/profile page.	Find Doctors	Ask for Appointment Help
Partner page	Healthcare provider wants onboarding.	Partner enquiry form.	Partner With Us	Talk to MarizFirst

4. Patient Discovery and Booking Flow

This is the core conversion flow and should be built before advanced dashboards or growth features. It must work smoothly on mobile, because many healthcare appointment users will search from phones.

Step	Screen/System Action	User/System Decision	Output
1	Patient lands on home, listing, city, speciality, blog, hospital, clinic, or doctor page.	Does user already know the provider?	User either searches or opens a profile directly.
2	Patient uses search or listing filters.	Search has matching results?	Show relevant doctors/hospitals/clinics or fallback suggestions.
3	Patient opens profile.	Profile active, verified, and published?	Display details, trust signals, slots, CTA, FAQs, disclaimer.
4	Patient selects preferred slot.	Slot AVAILABLE and capacity remains?	Proceed to short booking form.
5	Patient submits form.	Name, phone, city, slot, consent valid?	Create patient, appointment, lead, notification jobs, audit log.
6	System shows confirmation screen.	Appointment initial status must be Pending Confirmation.	Patient understands next step and receives WhatsApp acknowledgement.
7	Admin/partner acts on request.	Confirm, reschedule, cancel, or mark lost?	Patient receives final WhatsApp update.
8	Appointment date passes.	Was patient visit completed?	Admin/partner marks Completed or No Show.

4.1 Patient Booking Sequence

1. Search or navigate to a healthcare profile.
2. Review provider details: speciality, qualification, experience, fee where available, location, languages, registration number if available, facilities, and FAQs.
3. Select a visible preferred slot. The UI must clearly say that the selected slot is a request, not final confirmation.
4. Submit short form: name, mobile, city, preferred slot, consent, optional concern, optional email.
5. See success screen: request received, Pending Confirmation, WhatsApp update expected.
6. Receive WhatsApp acknowledgement.
7. Receive confirmed/rescheduled/cancelled update after admin or partner action.

4.2 Patient Form Requirements

Field	Required	Validation	UX Rule
Name	Yes	2-80 characters; no numbers-only input.	Visible label, clear error, no placeholder-only label.
Mobile	Yes	Indian mobile format initially; future international support optional.	Show country code pattern if used. Do not send to public analytics.
City	Yes	Must match supported city or allow typed fallback if	Pre-fill from search/profile when possible.

Field	Required	Validation	UX Rule
Preferred slot	Yes	configured. Must be active, not expired, capacity available.	Display selected date/time near submit button.
Concern	Optional	Max length; no HTML; stored securely.	Do not make this feel mandatory. Do not send to GA4.
Email	Optional	Valid email format.	Useful for future patient account/history, not mandatory.
Consent	Yes	Checkbox must be checked.	Explain contact/appointment purpose clearly.

5. Doctor Profile Flow

Screen Area	Content/Action	Flow Requirement
Profile hero	Doctor name, speciality, verified marker, city/area, primary CTA.	CTA visible above fold on desktop and sticky on mobile.
Professional details	Qualification, experience, registration number if available, languages, gender.	Structured and readable; avoid vague claims.
Fee/location block	Consultation fee if available, clinic/hospital address, map link.	No hidden fees or misleading copy.
Slot selector	Available dates/times.	Must show Pending Confirmation message before form.
Booking form	Short patient request form.	No login required in MVP.
FAQs	Profile-specific FAQs.	Enable FAQ schema only when FAQs exist and are accurate.
Disclaimer/trust	Appointment assistance, not emergency care or medical advice.	Visible near booking area and footer/legal links.

6. Hospital and Clinic Flow

Flow	Hospital Requirement	Clinic Requirement	User Outcome
Discovery	Show hospitals by city, departments, facilities, linked doctors.	Show clinics by location, services, doctors/team, availability.	User understands whether the organization matches their care need.
Profile review	Gallery, address, facilities, specialities, linked doctors, appointment CTA.	Address, services, team, facilities, local trust signals, booking CTA.	User can choose organization or a linked provider.
Booking path	Request appointment/enquiry for hospital or linked doctor.	Request appointment for clinic/team/slot.	Appointment created with correct owner type and route source.
Admin routing	Admin sees hospital/department/doctor context.	Admin sees clinic/team/slot context.	Request can be routed without confusion.
Partner action	Hospital partner sees scoped requests.	Clinic partner sees scoped requests.	Partner cannot see unrelated patient data.

7. Appointment State Machine

Non-Negotiable State Rule

Every patient-facing appointment starts as Pending Confirmation. Only admin or an authorized partner action can change it to Confirmed. The UI must never imply final confirmation before this action.

Status	Meaning	Allowed Next Statuses	Required Flow Action
PENDING_CONFIRMATION	Patient request submitted; not final.	CONFIRMED, RESCHEDULED, CANCELLED, LOST	Notify patient/admin/partner; show SLA timer in admin.

Status	Meaning	Allowed Next Statuses	Required Flow Action
CONFIRMED	Admin or partner approved final appointment.	COMPLETED, CANCELLED, RESCHEDULED, NO_SHOW	Send confirmation message with date, time, address, support contact.
RESCHEDULED	Original requested slot changed.	CONFIRMED, CANCELLED, LOST	Store old/new slot and reason; notify patient and partner.
CANCELLED	Appointment cancelled by patient, admin, or partner.	Closed; new request if needed.	Reason recommended; release slot capacity based on policy.
COMPLETED	Patient visit completed.	Closed	Used for partner quality, reports, commission fields.
NO_SHOW	Patient did not attend.	Closed unless admin reopens manually.	Reason/notes recommended.
LOST	Lead could not be converted or confirmed.	Closed	Reason required for operations reporting.

8. Slot Selection and Capacity Flow

Slot Status	Public Visibility	Selectable?	System Rule
AVAILABLE	Visible	Yes, if capacity remains.	Can be selected by patient. Must validate again at submission.
HOLD	Usually hidden or marked temporarily unavailable.	No, unless policy allows multi-capacity holds.	Optional strategy for pending requests; expires after configured time.
BOOKED	Hidden or marked full.	No	Capacity full or confirmed; not selectable.
BLOCKED	Hidden.	No	Admin/partner marked unavailable.
EXPIRED	Hidden.	No	Past slot; retained for logs.

8.1 Capacity Decision Rules

- Validate capacity twice: once while rendering slots and again during appointment creation.
- Appointment creation must be transactional to avoid duplicate/inconsistent booking states.
- For single-capacity slots, decide whether pending appointments place the slot on HOLD or only increment bookedCount after confirmation. HOLD is safer for trust; capacity-count strategy is better for high-volume clinics.
- Past slots must automatically become EXPIRED and unavailable.
- Admin must be able to block slots and resolve conflicts manually.

9. WhatsApp and Notification Flow

Trigger	Recipient	Message Purpose	Failure Handling
Appointment submitted	Patient	Acknowledge request received; status Pending Confirmation.	Log failure; show retry in admin; do not lose appointment.
Appointment submitted	Admin	Alert new request with source/profile/slot.	Show in dashboard regardless of external failure.
Appointment submitted	Partner	Alert scoped partner with accept/reject/reschedule actions.	Fallback to admin manual follow-up if partner notification fails.
Appointment confirmed	Patient	Send final date/time/address/provider instructions.	Retry and surface failure.
Appointment rescheduled	Patient + partner	Communicate old slot, new slot, reason if available.	Retry and log.
Appointment cancelled	Patient + partner	Communicate cancellation and support option.	Retry and log.
Reminder due	Patient	Remind 12-24 hours before confirmed appointment.	Queue retry; do not send for cancelled/lost/no-show.

- Use official WhatsApp Business API only. Do not automate WhatsApp Web for production messaging.
- Store providerMessageId, sentAt, status, failure reason, retry count, and webhook payload where useful.

- Webhook processing must verify provider signature/secret and be idempotent.
- Templates must avoid medical advice, treatment guarantees, emergency-care claims, and discount-led positioning.

10. Admin Operations Flow

Admin flow is the operational backbone. If admin cannot see, filter, act, and recover appointments quickly, the product will fail even if the public website looks strong.

Admin Module	Primary Flow	Important Rules
Dashboard	View appointment totals, pending SLA warnings, lead sources, top partners, notification failures.	Dashboard must prioritize Pending Confirmation and failed notifications.
Appointment inbox	Filter by status/date/doctor/hospital/clinic/city/source; open detail; confirm/reschedule/cancel/complete/no-show/lost.	Every status action must validate transition and create audit log.
Appointment detail	View patient info, selected slot, partner, source, UTM, notes, notification timeline, audit timeline.	Do not expose sensitive data to unauthorized roles.
Doctors	Create/edit/verify/publish doctor profiles, SEO, FAQs, media, active status.	Cannot publish incomplete critical profile fields.
Hospitals/Clinics	Manage org profile, facilities, linked doctors/team, gallery, location, CTA behavior.	Public visibility controlled by status.
Slots	View/block/create slots, resolve conflicts, bulk upload if needed.	Past/conflicting slots must be prevented.
Partners	Create accounts, assign role, manage plan/status/permissions.	Partner access must be scoped to assigned organization/profile.
Content/SEO	Publish blogs, city/speciality content, FAQs, legal pages, redirects.	Medical disclaimer review required for healthcare content.
Audit logs	Search sensitive actions and exports.	Append-only; normal admins cannot delete logs.

10.1 Admin Appointment Handling Flow

- Open Admin Dashboard and review Pending Confirmation count and SLA warnings.
- Open appointment detail and review patient, profile, slot, source, and partner response status.
- If partner accepted, confirm appointment and trigger patient confirmation.
- If partner rejected or no response, reschedule or call partner manually.
- If patient cannot be reached or slot unavailable, mark Cancelled or Lost with reason.
- After appointment date, mark Completed or No Show.
- Review reports for partner reliability and source quality.

11. Partner Dashboard Flow

Partner Screen	Flow	Guardrail
Login	Partner authenticates and lands on scoped dashboard.	Redirect by role; no admin/global data access.
Profile	View/edit own profile fields. Sensitive fields may require admin review.	Partner cannot publish fake claims or change verification without approval.
Slots	Add single/bulk slots, block unavailable dates, manage capacity.	Prevent past slots, conflicts, and over-capacity.
Appointments	View own appointment requests, accept/reject/request reschedule, add notes.	Ownership check required before query/action.
Patient detail	See only appointment-specific patient data after request.	No unrelated leads or bulk export unless permissioned.
Reports	View period-based pending/confirmed/completed/no-show counts.	P1: not needed before booking operations work.
Support request	Ask MarizFirst admin for profile/content/operations help.	P1: useful for scale but can be manual initially.

12. Content, SEO, and Publishing Flow

Content Type	Creation Flow	Publish Gate	Post-Publish Action
Doctor profile	Admin creates profile, adds core details, SEO, FAQs, media, verification.	Required fields complete; no fake ratings/reviews; status active.	Profile appears in sitemap and internal links.
Hospital/Clinic profile	Admin adds facility details, address, linked doctors/team, gallery, schema fields.	Location/facilities/CTA complete; status active.	Profile appears in relevant listing/city pages.
City/Speciality page	Content editor adds intro, FAQs, metadata, internal links.	No thin duplicate content; no medical claims.	Sitemap updates; internal links from relevant profiles.
Blog post	Draft -> review -> publish.	No diagnosis/treatment guarantee; metadata complete.	Blog indexed and linked to relevant speciality/city pages.
Legal page	Admin/super admin edits protected content.	Legal/compliance approval.	Footer link remains visible and crawlable.
Redirect	Admin adds source/destination/status.	Avoid redirect chains and loops.	Slug changes preserve SEO value.

13. Auth, RBAC, and Access Flows

Flow	Required Checks	Failure Behavior
Admin login	Valid credentials/session, admin role, secure cookie.	Deny access, log failed attempt if applicable.
Partner login	Valid partner role and assigned profile/organization.	Redirect to partner dashboard or deny if inactive.
Content editor login	Valid content role.	Hide patient/finance/appointment-sensitive screens.
Admin API request	Session + role permission before data access.	Return 403; audit sensitive denied attempts where useful.
Partner API request	Session + ownership-scoped query.	Return 404/403 without revealing existence of unrelated data.
Export action	Explicit permission + audit log.	Block export and show permission error.
Super admin dangerous action	Role + confirmation step.	Reject without confirmation; log action.

14. Search, Filters, Empty States, and Fallbacks

Scenario	Expected Flow	Fallback / Next Best Action
Search has exact matches	Show grouped or ranked results with type labels.	Allow filters and profile click.
Search has partial matches	Show closest city/speciality/provider matches.	Suggest related terms and WhatsApp support.
No results	Show no-results screen, not blank page.	Suggest nearby city/speciality, clear filters, contact support.
Filter returns zero	Keep filter drawer usable and show reset action.	Reset filters or broaden search.
Location unavailable	Do not block search.	Ask user to select city manually.
Search service error	Show safe error message.	Retry action and WhatsApp support CTA.

15. Error, Loading, Success, and Recovery States

State Type	Where Required	Rule
Loading	Search, listings, profiles, slot picker, dashboards, tables.	Use skeletons/spinners with stable layout; avoid blank screens.
Empty	No doctors, no slots, no appointments, no	Explain state and give next action.

State Type	Where Required	Rule
	notifications.	
Error	Form errors, API errors, payment-free booking errors, notification failure.	Show user-safe message; log technical detail server-side.
Success	Appointment submitted, status updated, profile saved, slot created.	Confirm action and show next step.
Pending	Appointment Pending Confirmation, partner response waiting.	Use explicit text, not color alone.
Blocked	Unauthorized route, inactive profile, blocked slot.	Show correct access/status message without leaking sensitive data.

16. Analytics and Conversion Tracking Flow

Event	Trigger	Properties	Privacy Rule
search_performed	User submits search.	query, city, speciality, result_count, page_location	No patient identifiers or concern text.
listing_filter_applied	User applies filters.	filter_type, filter_value, page_type	No personal data.
profile_viewed	Profile page viewed.	profile_id, profile_type, city, speciality, source_page	No patient identifiers.
slot_selected	User selects slot.	slot_id, profile_id, profile_type, date, time	No patient identifiers.
appointment_form_started	User starts form.	profile_id, source_page	Do not send typed field values.
appointment_submitted	Appointment request created.	appointment_id, profile_id, source_page, status	Do not send phone/email/concern to GA4.
whatsapp_cta_clicked	WhatsApp support clicked.	page_type, profile_id, location	No phone in event payload.
appointment_confirmed	Admin/partner confirms.	appointment_id, partner_id, time_to_confirm	Internal analytics only where needed.
appointment_rescheduled/cancelled	Status updated.	reason category, appointment_id, actor_role	Avoid sensitive notes.

17. Security and Privacy Flow Controls

Risk	Flow Control	Acceptance Criteria
Patient data exposure	Patient data only appears in admin and relevant partner appointment detail.	Unrelated partner URL/API manipulation cannot reveal data.
Spam appointments	Rate limit booking/search/auth; add CAPTCHA/Turnstile if needed.	Repeated spam submissions are throttled or challenged.
Unsafe healthcare claims	Content review gate before publish.	No emergency-care, guaranteed outcome, or medical diagnosis claims.
Unauthorized status changes	RBAC + transition validation + audit log.	Only permitted roles can change appointment state.
Webhook spoofing	Verify WhatsApp webhook signature/secret.	Invalid webhook rejected and logged.
Sensitive analytics leakage	Strip phone/email/concern from analytics payloads.	No sensitive fields visible in GA4 or public logs.
Destructive data loss	Soft delete business records; backups enabled.	Records recoverable and audit history preserved.

18. Screen Inventory and Route-Level Flow

Route / Screen	User Type	Primary Action	Connected Next Screens
/	Public	Search healthcare providers, open categories, book/WhatsApp CTA.	/doctors, /hospitals, /clinics, profile pages, /partner-with-us
/doctors	Public	Filter/search doctor listing.	/doctor/[slug], /doctors/[city], /doctors/[speciality]/[city]
/doctor/[slug]	Public	Review doctor, select slot, submit booking.	Booking success, WhatsApp, related profiles.

Route / Screen	User Type	Primary Action	Connected Next Screens
/hospitals	Public	Browse hospitals.	/hospital/[slug]
/hospital/[slug]	Public	Review facilities/linked doctors/request appointment.	Linked doctor profile, booking success.
/clinics	Public	Browse clinics.	/clinic/[slug]
/clinic/[slug]	Public	Review clinic/team/request appointment.	Booking success.
/specialities/[slug]	Public/SEO	Learn and find relevant providers.	Filtered listings/profile pages.
/cities/[slug]	Public/SEO	City healthcare hub.	Speciality pages/profile listings.
/blog/[slug]	Public/SEO	Read education content.	Relevant city/speciality/profile CTA.
/partner-with-us	Partner prospect	Submit partner enquiry.	Admin lead/review flow.
/admin	Admin	View operations dashboard.	Appointments, profiles, partners, notifications, audit logs.
/admin/appointments	Admin	Manage appointment inbox/statuses.	Appointment detail, notifications, audit timeline.
/partner	Partner	View scoped operations.	Profile, slots, appointments.

19. Edge Cases and Decision Rules

Edge Case	System Response	Owner
Patient submits same phone and slot repeatedly.	Throttle or merge duplicate request; show existing pending status if possible.	Backend/Admin
Slot becomes unavailable after page load.	Revalidate on submit; show slot unavailable and ask to choose another.	Backend/Frontend
Partner rejects request.	Admin offers reschedule or cancellation; patient receives update.	Partner/Admin
Partner does not respond within SLA.	Admin dashboard highlights breach; admin follows up manually.	Admin
WhatsApp message fails.	Log failure, show retry in admin, use manual fallback if urgent.	Backend/Admin
Profile unpublished after appointment request.	Existing appointment remains operational; public profile hidden.	Admin
Content editor tries to access patient data.	403 access denied; no data returned.	Security
Expired slot selected from stale browser.	Reject on submit and refresh slot list.	Backend
No doctors for city/speciality.	Show empty state with related specialities/cities and WhatsApp support.	Frontend/SEO
Patient asks emergency care.	Show disclaimer and emergency-care guidance language; do not claim emergency service.	Content/Admin

20. Acceptance Criteria and Flow QA Checklist

Area	Acceptance Criteria
Patient booking	User can search, open profile, select slot, submit form, and see Pending Confirmation success state.
Admin confirmation	Admin receives request and can confirm/reschedule/cancel with notification and audit log.
Partner scoped flow	Partner can only see own appointments and can accept/reject/reschedule requests.
WhatsApp flow	Acknowledgement, confirmation, reschedule, cancellation, reminder flows are logged and retryable.
SEO flow	Published profiles/pages generate metadata, schema, canonical, sitemap entries, and internal links.
Security flow	RBAC, ownership checks, rate limits, webhook verification, and sensitive analytics rules are enforced.
Accessibility flow	Search, filters, forms, drawers, dashboard actions work by keyboard and visible focus states.
Responsive flow	360/390/430 mobile, tablet, 1366/1440/1920 desktop screens have no broken layout or hidden CTAs.

Area	Acceptance Criteria
Error recovery	No result, API error, failed notification, unauthorized access, and stale slot states show useful recovery actions.
Operational readiness	Admin SOP exists for pending appointments, partner follow-up, reschedule/cancel, and failed notifications.

21. Screen-by-Screen Micro-Flows

This section breaks the product into implementation-level screen flows. Each screen must define entry condition, user action, validation, output, analytics, and failure behavior. This prevents vague handoff between design, frontend, backend, QA, and operations.

Screen	Entry Condition	Primary Interaction	Validation / System Check	Exit Condition
Home	Public user or SEO visitor lands on root route.	Search by doctor, speciality, hospital, clinic, city, or area.	Sanitize query; support empty state; track search_performed.	User moves to listing, profile, or support CTA.
Doctors Listing	User arrives from nav, search, city page, or speciality page.	Apply filters, compare cards, open profile.	Only active/published doctors; filters are canonical-safe.	User opens doctor profile or resets filters.
Doctor Profile	Doctor is active/published.	Review profile, choose slot, open form.	Profile data complete; slot availability fetched safely.	User submits appointment or clicks WhatsApp.
Slot Picker	User is on profile with slots available.	Select date/time.	Slot not expired, not blocked, capacity remains.	Selected slot populates form.
Booking Form	User selected profile and preferred slot.	Enter name, phone, city, optional concern/email, consent.	Server validates all fields, rate limits, slot capacity, consent.	Appointment created as Pending Confirmation.
Booking Success	Appointment created successfully.	Read next step, WhatsApp expectation, support options.	Status is Pending Confirmation; acknowledgement notification queued.	User can return home, WhatsApp support, or view related providers.
Admin Dashboard	Admin authenticated.	Review pending, SLA warnings, notification failures.	RBAC admin; aggregate queries permissioned.	Admin opens appointment/detail/listing modules.
Admin Appointment Detail	Admin opens a request.	Confirm, reschedule, cancel, complete, no-show, lost, add notes.	Allowed transition, notification queue, audit log.	Status updates and stakeholders notified.
Partner Dashboard	Partner authenticated.	View scoped requests, slots, profile.	Ownership-scoped query; inactive partners restricted.	Partner acts on own appointments only.
Partner Slot Manager	Partner/admin enters slot module.	Create, edit, block, bulk-create slots.	No past/conflicting slots; valid owner/capacity.	Slots become visible or hidden based on status.
Content Editor	Content/SEO user authenticated.	Create blog/FAQ/city/speciality content.	No medical guarantees; metadata required before publish.	Content published and sitemap updated.

22. Mobile App-Like Web Flow Rules

The MVP is web-first, but the patient experience should feel app-like on mobile. The flow must be optimized for thumb interaction, quick scanning, sticky actions, and minimal typing.

Mobile Area	Required Behavior	Bad Practice to Avoid
Header	Compact logo, search access, WhatsApp/help action, no overcrowded nav.	Desktop-style mega menu squeezed into mobile.
Search	Single prominent search box; city/speciality suggestions; recent/quick categories optional.	Multiple competing search boxes or hidden search.
Filters	Drawer with Apply, Reset, Close; focus management and body scroll lock.	Permanent sidebar, tiny checkboxes, no reset.
Doctor cards	Readable hierarchy: name, speciality, location, fee, availability, CTA.	Too many badges, tiny text, poor spacing.
Profile CTA	Sticky bottom bar with Book Appointment and WhatsApp Support.	CTA only at bottom after long scrolling.
Slot picker	Horizontal date chips or compact calendar; disabled states clear.	Tiny calendar grid that causes mis-taps.
Booking form	One-column, visible labels, large fields, inline validation.	Long multi-step form for MVP or forced login.

Mobile Area	Required Behavior	Bad Practice to Avoid
Success state	Clear Pending Confirmation message and next step.	Generic thank-you page without status clarity.
Dashboard mobile	Admin/partner tables must convert to cards where needed.	Wide tables causing horizontal scroll for core actions.

23. Admin Status Action Modals

Action	Modal Inputs	System Checks	Notifications	Audit Requirement
Confirm	Optional note, confirmed date/time/address if changed.	Appointment in allowed state; slot valid; actor permission.	Patient confirmation; partner update if admin confirmed.	Old/new status, actor, timestamp, note.
Reschedule	New slot/date/time, reason, optional note.	New slot available; old appointment not closed.	Patient + partner reschedule update.	Old slot, new slot, reason.
Cancel	Reason category, optional note.	Allowed actor and state.	Patient + partner cancellation.	Reason required/recommended.
Complete	Visit completed flag, optional internal note.	Appointment date reached or admin override.	Optional partner/admin record only.	Closed status log.
No Show	No-show reason/note.	Appointment previously confirmed.	Optional internal/partner update.	No-show status and actor.
Lost	Lost reason category.	Usually from pending/rescheduled states.	Optional patient support message if appropriate.	Reason required for reporting.
Retry Notification	Select failed notification.	Provider status failure exists; actor permission.	Retry same template or manual fallback.	Retry count and actor.

24. Partner Slot Management Details

Partner Task	Flow Steps	Validation	Admin Visibility
Create single slot	Open slots -> choose date/time/duration/capacity -> save.	No past time; capacity positive; no conflict for same owner.	Admin can view slot and appointment impact.
Bulk create slots	Choose date range, days, time windows, duration, capacity -> preview -> confirm.	Reject impossible ranges, duplicate conflicts, invalid capacities.	Bulk creation appears in audit/activity.
Block slot/date	Select slot/date -> block with reason.	Cannot block confirmed appointment without admin handling.	Admin sees blocked reason and affected requests.
Edit slot	Change time/capacity/status.	Prevent reducing capacity below booked/pending count unless admin override.	Audit log captures before/after.
Respond to appointment	Open request -> accept/reject/reschedule.	Ownership check; allowed transition.	Admin timeline updates instantly.
Mark completed/no-show	Open confirmed appointment after visit time -> mark outcome.	Only appointment-specific data visible.	Reports and partner quality metrics update.

25. Data Touchpoints Across Flow

Flow Moment	Data Created/Read	Sensitive?	Storage / Access Rule
Profile view	Doctor/hospital/clinic/city/speciality public data.	No, if public profile fields only.	Only published active records shown.
Slot selection	AppointmentSlot id, owner, start/end, capacity/status.	Operational data.	Validate again on submit.
Form start	Profile/source context, no typed values in analytics.	Potentially sensitive if typed values captured.	Do not store until submit except safe analytics event.
Form submit	Patient name, phone, city, optional email/concern, consent.	Yes.	Restricted admin/scoped partner access.
Appointment create	Appointment, Lead, Notification jobs, AuditLog.	Yes.	Transactional creation; no public exposure.
Admin status update	Status, notes, actor, timestamp, notification.	Yes.	Audit log required.
Partner action	Scoped appointment response	Yes.	Ownership-scoped access only.

Flow Moment	Data Created/Read	Sensitive?	Storage / Access Rule
Analytics event	and partner notes. Funnel metadata, profile ids, status categories.	Should not be sensitive.	No phone, email, concern, or health notes.
SEO publish	Public content, metadata, schema, sitemap.	No, if reviewed.	No patient data or private notes in public pages.

26. Operational SOP Flows

SOP	Trigger	Steps	Success Output
Pending appointment handling	New appointment or SLA warning.	Review request -> contact partner if no response -> confirm/reschedule/cancel -> notify patient.	Patient receives clear final update.
Failed notification handling	Notification status failed or webhook error.	Open failure list -> retry -> if still failed, use manual WhatsApp/call fallback -> log note.	No silent communication failure.
Partner onboarding	New doctor/hospital/clinic interested.	Collect profile data -> verify -> create partner account -> add slots -> publish profile.	Partner appears in public discovery and scoped dashboard.
Profile quality review	Before publish or periodic audit.	Check required fields, claims, media, location, SEO metadata, disclaimers.	Profile is accurate, complete, and SEO-ready.
SLA breach escalation	Pending request approaches 15-30 minute SLA.	Admin gets warning -> call partner -> update status -> log note.	Operational trust preserved.
Incorrect appointment data	Patient/partner reports wrong details.	Admin edits/reschedules/cancels -> notification sent -> audit log.	Correct information reaches patient.
Data deletion request	Patient requests deletion process.	Verify request -> follow legal/data policy -> update/delete/anonymize where appropriate -> log action.	Request handled without breaking audit/compliance needs.

27. UAT Test Scenarios

Scenario ID	Scenario	Expected Result
UAT-01	Patient searches cardiologist in launch city, opens profile, selects slot, submits valid form.	Appointment created as Pending Confirmation; WhatsApp acknowledgement queued; admin/partner notified.
UAT-02	Patient submits invalid phone or no consent.	Form does not submit; accessible error message shown; no appointment created.
UAT-03	Two users submit same single-capacity slot at nearly same time.	System prevents overcapacity; one request succeeds based on final slot strategy.
UAT-04	Admin confirms pending appointment.	Status becomes Confirmed; patient notification queued; audit log written.
UAT-05	Admin reschedules appointment.	Old/new slot stored; patient and partner receive update; audit log written.
UAT-06	Partner logs in and tries to open another partner appointment URL.	Access denied or not found; no sensitive data exposed.
UAT-07	Content editor logs in and tries to access patient appointment data.	Access denied; navigation does not show restricted modules.
UAT-08	WhatsApp provider fails on confirmation message.	Failure visible in admin notification dashboard with retry option.
UAT-09	Published doctor profile is checked for SEO.	Metadata, schema, canonical, internal links, and sitemap entry exist.
UAT-10	Mobile booking on 390px width.	No horizontal scroll; sticky CTA works; form labels/errors readable; tap targets usable.
UAT-11	No search results for selected city/speciality.	Helpful empty state shown with related suggestions and WhatsApp support.
UAT-12	Expired slot is opened from stale browser tab.	Submit rejected; user asked to select another available slot.

28. Release Readiness Flow Checklist

Release Gate	Must Pass Before Launch
Public discovery	Home, listings, profile pages, city/speciality pages, hospital/clinic pages, blog, legal pages are responsive and linked.
Booking conversion	Search -> profile -> slot -> form -> Pending Confirmation works without forced login.
Admin operations	Admin can manage appointments, profiles, partners, content, SEO, notifications, and audit logs.
Partner workflow	Partner can manage only own profile, slots, and appointment requests.
Notifications	Acknowledgement, admin alert, partner alert, confirmation, reschedule, cancellation, reminder paths are tested.
SEO	Dynamic metadata, sitemap, robots, schema, canonical, internal linking, alt text, and redirects are checked.
Security	RBAC, ownership checks, rate limits, validation, secure cookies, webhook verification, and backups are active.
Accessibility	Keyboard navigation, visible focus, labels, contrast, form errors, drawers/modals, tap targets pass QA.
Performance	Mobile LCP/CLS/INP targets are addressed; images compressed; listing pagination/search indexes ready.
Operations	Admin SOPs, partner onboarding checklist, failed notification fallback, SLA process, and first-week monitoring are ready.

Appendix A: Minimum MVP Flow Release Gate

- Public route flow works: Home -> Search/Listing -> Profile -> Slot -> Booking Form -> Pending Confirmation.
- Admin route flow works: Login -> Dashboard -> Appointment Inbox -> Status Action -> Notification -> Audit Log.
- Partner route flow works: Login -> Own Appointments -> Accept/Reject/Reschedule -> Notification -> Audit Log.
- SEO route flow works: Published profile/city/speciality/blog pages appear in sitemap with correct metadata and schema.
- Security route flow works: unauthorized users and unrelated partners cannot access protected records.
- Notification route flow works: WhatsApp messages are queued, sent, tracked, and retryable.

Appendix B: Flow Ownership Matrix

Flow Area	Product Owner	Engineering Owner	QA Focus
Patient discovery and booking	Product + UX	Frontend + Backend	Conversion, validation, mobile UX.
Appointment state machine	Product + Operations	Backend	Allowed transitions, audit logs, notifications.
Admin operations	Operations	Backend + Frontend	Filters, status actions, SLA, recovery.
Partner dashboard	Partnerships + Product	Backend + Frontend	Scoped access, slot management, action accuracy.
SEO publishing	SEO + Content	Frontend + Backend	Metadata, sitemap, schema, internal links.
Security and privacy	Security + Product	Backend + DevOps	RBAC, ownership, rate limits, logging.
Analytics	Growth + Product	Frontend + Backend	Event accuracy and privacy-safe payloads.